# Checklist for the functionality of the products

The list of features that will be checked during the final demonstration are listed below:

## IPOS-SA sub-system

### *IPOS-SA-ACC package*

- creating/changing/deleting a user account
  - ability to create new user accounts (login/password) and assign roles (access privileges)
  - users should be allowed to do only things defined by their access privileges and not allowed to do things which exceed their privileges.
  - change roles (promote/demote) users to different level of access.
    - This can be done via deleting an existing user account and creating it with different privileges.
- creating/changing/deleting a merchant account
- setting up a credit limit and a discount plan for a merchant account
- maintaining account state (from “normal” to “suspended” and eventually to “in default”) including reactivating accounts “in default” to “normal”

### *IPOS-SA-CAT package*

- adding/deleting/modifying a product in the catalogue
  - setting/changing minimum stock availability.
- search in the catalogue by product ID, keyword, etc.
- adding new stock (i.e. increasing stock availability)
- warnings for stock below the set lower bound (e.g. when a stock item is open for modification, or when a user logs in)

### *IPOS-SA-ORD package*

- accepting new orders (submitted from IPOS CA),
  - generating invoices and storing them on SA.
- managing the order status (from “accepted” to “ready to dispatch”, then to “dispatched”, then to delivered).
- monitoring current balance of a merchant.
- Recording payments from merchants using various forms of payment (e.g. bank transfers).
  - Recording payments may change the merchant account status from “suspended” to “normal” provided the outstanding balance from previous month has been cleared.
- observing the list of orders taken but not completed.

### *IPOS-SA-RPT package*

- ability to generate all reports required with an option to print them.
  - generating a file (e.g. a pdf file) will be considered equivalent to printing.
- generating reminders to merchant debtors (shown on CA screen) every time a merchant debtor tries to log in.

## IPOS-CA sub-system

### *IPOS-CA-ORD package*

- placing new orders with SA
- checking progress with SA order
- search of SA orders and viewing invoices.
- Check status of own merchant account.
- (optional) record merchant account details on CA for ease of access to SA by different users of IPOS - CA.

### *IPOS-CA-USER package*

- create/change/delete an IPOS-CA user account with different privileges.

### *IPOS-CA-CUST package*

- create/change/delete an IPOS-CA customer account.
  - set up/modify a credit limit.
  - set up/modify a discount plan for an existing account holder (customer).
- generate reminders to debtors (1st and 2nd reminders)
- maintain customer account (from "normal" to "suspended" and then to "in default") including reactivation of an account "in default" by an authorised user, provided the debt from the previous month has been cleared.

### *IPOS-CA-Templates package*

- Ability to create/modify reminder templates (1st and 2nd reminder).
  - Make sure that the templates are used in any future reminders.

### *IPOS-CA-Stock package*

- dealing with local stock:
  - adding new stock (a new item and increasing the quantity of an existing stock item)
  - dealing with order deliveries (when an IPOS-SA order is delivered)[1]
  - listing (on request) all items, which are below their respective lower stock bounds.

### *IPOS-CA-Sales package*

- process purchases by customers, produce retail invoices.
- take payments from customers using various forms of payment (including cash and credit card)
- maintain orders received via the PU portal (order status will be changed from "accepted" to "ready for shipment", then to "shipped" and finally "delivered").

### *IPOS-CA-RPT package*

- ability to generate all reports required with option to print them.
  - generating a file (e.g. a pdf file) will be considered equivalent to printing.

## IPOS-PU sub-system

### *IPOS-PU-Members package*

- application for non-commercial account (membership)
- application for commercial account (membership)
- login/ by non-commercial members
- non-commercial customer can view their purchase history.
- ability by non-commercial members to track their orders (see the state of orders).

### *IPOS-PU-Sales package*

- show IPOS - CA catalogue.
  - show a list of links to active promotion campaigns (provided there is at least one active campaign at the time a customer is scanning the on-line catalogue)
  - clicking the link of an active promotion campaign will show details on promotion campaign.
- maintain a shopping cart for customers (and non-commercial members)
  - apply discounts to products included in active promotion campaigns.
  - maintain all counters envisaged in the initial statement of requirements (when links/product items are selected/added to cart/paid for).
- checkout
  - accounting for non-commercial customer getting a 10% discount from every tenth order.
  - Collect delivery address.
  - Collect online payment.
  - Send the customer a sale confirmation with an order tracking link.

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[1] An assumption can be made that all orders placed with IPOS-SA are delivered as placed (i.e. the delivered items and quantities are as specified in the order).

### *IPOS-PU-Comms package*

- (optional) Configure access to SMTP server.
- (optional) Configure access to an online payment processor.
- Sending mails on behalf of IPOS-SA
- Processing credit card payment requests on behalf of IPOS-CA (card payments are processed via the online payment processor)

### *IPOS-PU-PRM package*

- Create/modify/delete/terminate early promotion campaigns.
  - Setting period of campaign
  - Including items from IPOS-CA catalogue
  - Setting discount rates

### *IPOS-PU-RPT package*

- Create all reports envisaged in the initial requirements documents with option to print them.
  - generating a file (e.g. a pdf file) will be considered equivalent to printing.

## G: General functionality

- Sensible search
- Cascaded deletes/updates
- GUI – appearance of the forms (intuitive, easy to use) and consistency between the forms.

# Sample Data

The data provided below should be used to populate the databases of IPOS-SA, IPOS-CA, and IPOS – PU subsystems. Prior to populating the databases, you **MUST** remove all the data from the database of the respective subsystem, i.e. leave all tables blank, with a single exception – a single (default) login which will allow you to create the required login usernames/passwords.

# IPOS_SA subsystem

## Logins

The following user account must be active with IPOS-SA:

| Username | Password | role |
|---|---|---|
| Sysdba | London_weighting | Administrator |
| manager | Get_it_done | Director of Operations |
| accountant | Count_money | Senior accountant |
| clerk | Paperwork | Accountant |
| warehouse1 | Get_a_beer | Warehouse employee |
| warehouse2 | Lot_smell | Warehouse employee |
| delivery | Too_dark | Delivery department employee |

## Account details of merchants registered with InfoPharma

**Account Holder Name**: CityPharmacy
**Account No**: ACC0001
**Contact Name**: Prof David Rhind
**Address**: Northampton Square, London EC1V 0HB
**Phone**: 0207 040 8000
**Credit Limit**: £10,000
**Agreed Discount**: Fixed
**Discount Rate**: 3%
**login**: city
**password**: northampton

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**Account Holder Name**: Cosymed Ltd
**Account No**: ACC0002
**Contact Name**: Mr Alex Wright
**Address**: 25, Bond Street, London WC1V 8LS
**Phone**: 0207 321 8001
**Credit Limit**: £5,000
**Agreed Discount**: Variable (on volume per month)
**Discount Rate**:

| Volume | Discount Rate |
|---|---|
| < £1000 : | 0 % |
| £1000 - £2000: | 1% |
| £2000+ : | 2 % |

**login**: cosymed
**password**: bondstreet

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**Account Holder Name**: HelloPharmacy
**Account No**: ACC0003
**Contact Name**: Mr Bruno Wright
**Address**: 12, Bond Street, London WC1V 9NS
**Phone**: 0207 321 8002
**Credit Limit**: £5,000
**Agreed Discount**: Variable (on volume per month)
**Discount Rate**:

| Volume | Discount Rate |
|---|---|
| < £1000 : | 0 % |
| £1000 - £2000: | 1% |
| £2000+: 3 % | |

**login**: hello
**password**: there

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## InfoPharma catalogue

| Item ID | Description | Package Type | Unit | Units in a pack | Package Cost, £ | Availability, packs | Stock limit, packs |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 100 00001 | Paracetamol | box | Caps | 20 | 0.10 | 10,345 | 300 |
| 100 00002 | Aspirin | box | Caps | 20 | 0.50 | 12, 453 | 500 |
| 100 00003 | Analgin | box | Caps | 10 | 1.20 | 4,235 | 200 |
| 100 00004 | Celebrex, caps 100 mg | box | Caps | 10 | 10.00 | 3,420 | 200 |
| 100 00005 | Celebrex, caps 200 mg | box | caps | 10 | 18.50 | 1,450 | 150 |
| 100 00006 | Retin-A Tretin, 30 g | box | caps | 20 | 25.00 | 2,013 | 200 |
| 100 00007 | Lipitor TB, 20 mg | box | caps | 30 | 15.50 | 1,562 | 200 |
| 100 00008 | Claritin CR, 60g | box | caps | 20 | 19.50 | 2,540 | 200 |
| | | | | | | | |
| 200 00004 | Iodine tincture | bottle | ml | 100 | 0.30 | 2,2134 | 200 |
| 200 00005 | Rhynol | bottle | ml | 200 | 2.50 | 1,908 | 300 |
| | | | | | | | |
| 300 00001 | Ospen | box | caps | 20 | 10.50 | 809 | 200 |
| 300 00002 | Amopen | box | caps | 30 | 15.00 | 1340 | 300 |
| | | | | | | | |
| 400 00001 | Vitamin C | box | caps | 30 | 1.20 | 3,258 | 300 |
| 400 00002 | Vitamin B12 | box | caps | 30 | 1.30 | 2,673 | 300 |
| | | | | | | | |

# IPOS_CA subsystem

The following ***user*** account must be active with the version of IPOS-CA deployed at **Cosymed Ltd.**:

| Username | Password | Role |
|---|---|---|
| sysdba | masterkey | Administrator |
| manager | Get_it_done | Director of Operations/Manager |
| accountant | Count_money | Senior accountant |
| clerk | Paperwork | Accountant |

## Customers registered as account holders with Cosymed Ltd.

**Account Holder Name**: Ms Eva Bauyer
**Account No**: ACC0001
**Contact Name**: Ms Eva Bauyer
**Address**: 1, Liverpool street, London EC2V 8NS
**Phone**: 0207 321 8001
**Credit Limit**: £500
**Agreed Discount**: Fixed: 3%

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**Account Holder Name**: Mr Glynne Morrison
**Account No**: ACC0002
**Contact Name**: Ms Glynne Morisson
**Address**: 1, Liverpool street, London EC2V 8NS
**Phone**: 0207 321 8001
**Credit Limit**: £500
**Agreed Discount**: Variable (on volume per month)
**Discount Rate**:

| Volume | Discount Rate |
|---|---|
| < £100 : | 0 % |
| £100 - £300: | 1% |
| £300+ : | 2 % |

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## Stock availability at Cosymed Ltd

| Item ID | Description | Package Type | Unit | Units in a pack | Package Cost, £ | Availability, packs | Stock limit, packs |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 100 00001 | Paracetamol | Box | Caps | 20 | 0.10 | 121 | 10 |
| 100 00002 | Aspirin | Box | Caps | 20 | 0.50 | 201 | 15 |
| 100 00003 | Analgin | Box | Caps | 10 | 1.20 | 25 | 10 |
| 100 00004 | Celebrex, caps 100 mg | Box | Caps | 10 | 10.00 | 43 | 10 |
| 100 00005 | Celebrex, caps 200 mg | Box | Caps | 10 | 18.50 | 35 | 5 |
| 100 00006 | Retin-A Tretin, 30 g | Box | Caps | 20 | 25.00 | 28 | 10 |
| 100 00007 | Lipitor TB, 20 mg | Box | Caps | 30 | 15.50 | 10 | 10 |
| 100 00008 | Claritin CR, 60g | Box | Caps | 20 | 19.50 | 21 | 10 |
| | | | | | | | |
| 200 00004 | Iodine tincture | Bottle | Ml | 100 | 0.30 | 35 | 10 |
| 200 00005 | Rhynol | Bottle | Ml | 200 | 2.50 | 14 | 15 |
| | | | | | | | |
| 300 00001 | Ospen | Box | Caps | 20 | 10.50 | 78 | 10 |
| 300 00002 | Amopen | Box | Caps | 30 | 15.00 | 90 | 15 |
| | | | | | | | |
| 400 00001 | Vitamin C | Box | Caps | 30 | 1.20 | 22 | 15 |
| 400 00002 | Vitamin B12 | Box | Caps | 30 | 1.30 | 43 | 15 |
| | | | | | | | |

Retail prices are calculated with 100% mark up to the unit cost. 0.0% VAT is applied to the retail price.

# IPOS_PU subsystem

The following ***user*** accounts must be active with the version of IPOS-PU linked to merchant **Cosymed Ltd**:

| Username | Password | Role |
|---|---|---|
| sysdba | masterkey | Administrator |
| manager | GetPU_it_done | PU-Admin |

The following membership applications were made via IPOS-PU (Cosymed Ltd):

**Non-commercial Member**
**Account:** PU0001
**Email address**: cool@example.com[2]
**password**: 12ss_56_SS

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**Non-commercial Member**
**Account No**: PU0002
**Email address**: cool1@example.com[3]
**password**: 34pp_78_LL

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**Commercial Member**
**Account No**: PU0003
**Company Name**: Pond Pharmacy
**Address**: Chislehurst
25, High Street
BR7 5BN
**Company House Registration**: UK10003429CompH
**Email address**: pondPharma@example.com[4]

[2] You can amend this email to a valid email address, so that emails can be sent to it.
[3] You can amend this email to a valid email address, so that emails can be sent to it.
[4] You can amend this email to a valid email address, so that emails can be sent to it.

## Scenarios

1. On 20 February 2026 CityPharmacy, placed the following order with IPOS-SA:

| Item ID | Description | Quantity | Unit Cost, £ | Total, £ |
|---|---|---|---|---|
| 100 00001 | Paracetamol | 10 | 0.10 | 1.00 |
| 100 00003 | Analgin | 20 | 1.20 | 24.00 |
| 200 00004 | Iodine tincture | 20 | 0.30 | 3.60 |
| 200 00005 | Rhynol | 10 | 2.50 | 25.00 |
| 300 00001 | Ospen | 10 | 10.50 | 105.00 |
| 300 00002 | Amopen | 20 | 15.00 | 300.00 |
| 400 00001 | Vitamin C | 20 | 1.20 | 24.00 |
| 400 00002 | Vitamin B12 | 20 | 1.30 | 26.00 |
| | | **Grand Total:** | | **508.60** |

The goods were delivered on 23 February 2026, 15:00 to the merchant's premises by the InfoPharma's own courier service.

2. On 25 February 2026, the merchant Cosymed Ltd, placed the following order with IPOS-SA:

| Item ID | Description | Quantity | Unit Cost, £ | Total, £ |
|---|---|---|---|---|
| 100 00001 | Paracetamol | 10 | 0.10 | 1.00 |
| 100 00003 | Analgin | 20 | 1.20 | 24.00 |
| 200 00005 | Rhynol | 10 | 2.50 | 25.00 |
| 300 00002 | Amopen | 20 | 15.00 | 300.00 |
| 400 00002 | Vitamin B12 | 20 | 1.30 | 26.00 |
| | | **Grand Total:** | | **376.00** |

The goods were delivered on 26 February, 17:00 to the merchant's premises by the DHL.

3. On 25 February 2026, the merchant HelloPharmacy placed the following order with IPOS - SA:

| Item ID | Description | Quantity | Unit Cost, £ | Total, £ |
|---|---|---|---|---|
| 100 00003 | Analgin | 20 | 1.20 | 24.00 |
| 200 00004 | Iodine tincture | 20 | 0.30 | 3.60 |
| 300 00001 | Ospen | 3 | 10.50 | 31.50 |
| 300 00002 | Amopen | 10 | 15.00 | 150.00 |
| 400 00001 | Vitamin C | 20 | 1.20 | 24.00 |
| 400 00002 | Vitamin B12 | 20 | 1.30 | 26.00 |
| | | **Grand Total:** | | **259.10** |

The goods were delivered on 27 February 2026, 10:00 to the merchant's premises by the DHL.

4. On 10 March 2026, the merchant Cosymed Ltd placed the following order with IPOS - SA:

| Item ID | Description | Quantity | Unit Cost, £ | Total, £ |
|---|---|---|---|---|
| 200 00005 | Rhynol | 10 | 2.50 | 25.00 |
| 300 00001 | Ospen | 10 | 10.50 | 105.00 |
| 300 00002 | Amopen | 20 | 15.00 | 300.00 |
| | | **Grand Total:** | | **430.00** |

The goods were delivered on 12 March, 11:00 to the merchant's premises by the InfoPharma's own courier service.

5. On 25 March 2026, the merchant HelloPharmacy placed the following order with IPOS - SA:

| Item ID | Description | Quantity | Unit Cost, £ | Total, £ |
|---|---|---|---|---|
| 100 00003 | Analgin | 20 | 1.20 | 24.00 |
| 100 00004 | Celebrex, caps 100 mg | 5 | 10.00 | 50.00 |
| 100 00005 | Celebrex, caps 200 mg | 5 | 18.50 | 92.50 |
| 100 00006 | Retin-A Tretin, 30 g | 5 | 25.00 | 125.00 |
| 100 00007 | Lipitor TB, 20 mg | 10 | 15.50 | 155.00 |
| 300 00001 | Ospen | 10 | 10.50 | 105.00 |
| 300 00002 | Amopen | 20 | 15.00 | 300.00 |
| 400 00002 | Vitamin B12 | 20 | 1.30 | 26.00 |
| | | **Grand Total:** | | **877.50** |

The goods were delivered on 27 March, 10:00 to the merchant's premises by the InfoPharma's own courier service.

6. On 1 April 2026, the merchant HelloPharmacy placed the following order with IPOS - SA:

| Item ID | Description | Quantity | Unit Cost, £ | Total, £ |
|---|---|---|---|---|
| 100 00003 | Analgin | 20 | 1.20 | 24.00 |
| 100 00004 | Celebrex, caps 100 mg | 5 | 10.00 | 50.00 |
| 100 00005 | Celebrex, caps 200 mg | 5 | 18.50 | 92.50 |
| 100 00006 | Retin-A Tretin, 30 g | 5 | 25.00 | 125.00 |
| 100 00007 | Lipitor TB, 20 mg | 10 | 15.50 | 155.00 |
| 300 00001 | Ospen | 10 | 10.50 | 105.00 |
| 400 00002 | Vitamin B12 | 20 | 1.30 | 26.00 |
| | | **Grand Total:** | | **577.50** |

The goods were delivered on 3 April 2026, 10:00 to the merchant's premises by the InfoPharma's own courier service.

7. HelloPharmacy have made no payments since 5 March 2026 when they cleared their balance.

8. CityPharmacy have made a full payment on 15 March 2026 and cleared their balance using a bank transfer to the account of InfoPharma.

9. Cosymed Ltd have made a full payment on 15 March 2026 and cleared their balance using company's credit card.

10. On 1 March 2026 Eva Bauyer bought the following goods from Cosymed Ltd:

| | |
|---|---|
| Ospen | 1 Boxes |
| Vitamin C | 2 Boxes |
| Amopen | 2 Boxes |
| Vitamin B12 | 2 Boxes |

11. On 3 March 2026, the following purchases of goods were made from the pharmacy shops of Cosymed Ltd (i.e. via the IPOS-CA subsystem) by several cash customers:

a)

| | |
|---|---|
| Aspirin | 2 Boxes |
| Analgin | 3 Boxes |

For goods, the customer paid cash.

b)

| | |
|---|---|
| Celebrex, caps 100 mg | 2 Boxes |
| Retin-A Tretin, 30 g | 2 Boxes |

Paid with credit card Visa.

c)

| | |
|---|---|
| Lipitor TB, 20 mg | 1 Box |
| Claritin CR, 60g | 1 Box |

Paid cash.

d)

| | |
|---|---|
| Celebrex, caps 200 mg | 1 Box |
| Iodine tincture | 2 Bottles |
| Rhynol | 2 Bottles |

Paid with cash.

e)

| | |
|---|---|
| Ospen | 2 Boxes |
| Vitamin C | 2 Boxes |

Paid with debit card.

f)

| | |
|---|---|
| Amopen | 3 Boxes |
| Vitamin B12 | 2 Boxes |

Paid cash.

12. On 5 March 2026 Glynne Morisson bought from a Cosymed Ltd Pharmacy Shop Ltd the following goods:

| | |
|---|---|
| Aspirin | 2 Boxes |
| Analgin | 3 Boxes |
| Celebrex, caps 100 mg | 2 Boxes |
| Retin-A Tretin, 30 g | 2 Boxes |

The cost of goods was added to Glynne's account with Cosymed Ltd.

13. On 1 April Eva Bauyer bought the following goods from a Cosymed Pharmacy shop:

| | |
|---|---|
| Ospen | 1 Boxes |
| Analgin | 3 Boxes |
| Celebrex, caps 100 mg | 2 Boxes |
| Vitamin B12 | 2 Boxes |

The cost of goods was added to Eva's account with Cosymed Ltd.

14. Glynne Morisson paid on 29 March the full balance with a credit card.

15. The last payment from Eva Bauyer was received on 30 February.

16. On the 15 April reminding letters were generated and sent to all debtors. The same must be done on 29 April.

17. On the 15 March 2026, the Admin – PU user of IPOS-PU created a promotion campaign as follows:

| | |
|---|---|
| Name: | March Promotion |
| Valid from: | 15 March 2026 |
| Valid to: | 20 April 2026 |
| Item | Discount rate |
| Aspirin | 5% |
| Analgin | 10% |
| Celebrex, caps 100 mg | 10% |
| Retin-A Tretin, 30 g | 20% |

18. On the 30 March 2026, the Admin – PU user of IPOS-PU created another promotion campaign as follows:

| | |
|---|---|
| Name: | April Promotion |
| Valid from: | 5 April 2026 |
| Valid to: | 10 April 2026 |
| Item | Discount rate |
| Ospen | 20% |
| Vitamin C | 10% |

19. On the 20 of March 2026 Peter Popov scanned the Cosymed Ltd online catalogue (via the PU subsystem), clicked on the link of "March Promotion". Then from the list of the promotion campaigns, he added two items to the shopping cart:
- Aspirin - 1 box
- Retin-A Tretin, 30 g – 1 box.

He then paid using his AmEx credit card: 0000 000000 0000 0001, security code 3245, valid until 30/08/2030.
The payment has been successful. Peter was sent an email confirming the sale and with a link to track the delivery.

20. Between 1st and 20th of March 2026 the non-commercial customer PU0001 (see login details above) has made 8 online purchases of various goods (details of these have not been stored, but the number of purchases (8) has been stored by the PU). On the 8th of April 2026 PU0001 logged in again and checked the active promotion campaigns. Then they added Ospen – 1 box to the shopping cart. They checked out and paid for the goods using their Mastercard 0001 0002 0003 0005 with security code 032 and expiry date 9/2028.
The payment transaction has been successful and PU0001 received an email confirming the purchase with an order tracking link included in the email.